Minkyu Kim

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Education

Seoul National University **Seoul, South Korea**
M.S. in Department of Intelligence and Information *Sept 2024 – Current*
- Advisor: Prof. Nojun Kwak

B.S. in Mechanical Engineering & Interdisciplinary Major in Artificial Intelligence *Mar 2019 – Aug 2024*
- GPA: 3.86/4.3 (Major GPA: 3.90, Interdisciplinary Major GPA: 3.94)
- 1.5 years of absence due to obligatory military service (2021 - 2022)

Incheon Academy of Science and Arts **Incheon, South Korea**
- High school for talented students in science and mathematics *Mar 2016 – Feb 2019*

Research Experience

Multi-Input Fusion for Accelerated Diffusion Generation **Seoul, South Korea**
***M.S. Student**, Machine Intelligence & Pattern Analysis Lab* *Feb 2025 - Current*

Advisor: Prof. Nojun Kwak, Seoul National University and Prof. Jangho Kim, Kookmin University

- Proposed a novel framework that boosts diffusion throughput by fusing multiple inputs to be processed in a single model inference, offering a new efficiency dimension beyond step reduction.
- Developed and validated a scalable, diffusion-model–agnostic framework with a timestep-variant sampling strategy that significantly improves throughput with minimal quality loss.

Learning Framework without Human Priors **Seoul, South Korea**
***M.S. Student**, Machine Intelligence & Pattern Analysis Lab* *Jan 2025 - Current*

Advisor: Prof. Nojun Kwak, Seoul National University

- Proposed a framework that enables models to extrapolate without human priors by training on random labels while preserving pretrained structure.
- Designed the AX objectives, conducted theoretical analysis of SGD dynamics, and validated the method across SFDA and object discovery tasks.

Marine Safety Management Using Drones for Threat Detection **Seoul, South Korea**
***M.S. Student**, Machine Intelligence & Pattern Analysis Lab* *Sept 2024 - Dec 2024*

Advisor: Prof. Nojun Kwak, Seoul National University

- Proposed a drone-based marine safety management system for real-time detection of threatening marine life near humans using on-device AI.
- Developed a model that detects marine creatures and humans on horizontal view finetuning YOLO model and applied post-training quantization (PTQ) for on-device deployment.

Large VQA Models for Dermatological Diagnosis **Seoul, South Korea**
***Research Internship**, Machine Intelligence & Pattern Analysis Lab* *Jan 2024 - June 2024*

Advisor: Prof. Nojun Kwak, Seoul National University

- Developed multimodal medical VQA systems for dermatological diagnosis.
- Proposed a dermatological diagnosis model by fine-tuning large vision–language models on MEDIQA-M3G and RAG-based synthetic datasets, achieving 2nd place in the English section of the shared task.

Fault Detection Enhancement in Journal Bearings Using Generative Models **Seoul, South Korea**
***Undergraduate Researcher**, Hyperautomation Artificial Intelligence Lab* *Jan 2024 - Aug 2024*

Advisor: Prof. Byeng Dong Youn, Seoul National University

- Applied generative models to synthesize abnormal vibration data for journal bearings, improving fault detection performance under data imbalance conditions.
- Demonstrated the industrial applicability of AI in mechanical systems, leading to selection for the Outstanding BS Thesis Presentation Award.

Slope-Control Algorithm for Electric Walking Aids Using FSR Foot Sensors **Incheon, South Korea**
***High School Research Intern**, Motion Control Lab* *Mar 2017 – Dec 2017*

Advisor: Prof. Yoon Su Baek, Yonsei University

- Designed FSR-based foot sensors to calculate the Center of Pressure (CoP) and predict ground slope and user intent during walking.
- Developed a control algorithm for electric walking aids that adjusts speed and assistance on inclines, enhancing safety and mobility for elderly users.

Design of Automatic Retractable Dome Structure Using Petal Arrangement Patterns **Incheon, South Korea**
***High School Student**, Incheon Academy of Science and Arts* *Aug 2016 – Mar 2017*

Advisor: T. Deokhan Yun

- Designed a dome structure inspired by the blooming pattern of the Mugunghwa (Hibiscus syriacus), the national flower of Korea.
- Selected as a **Korean national representative** and competed in the Intel ISEF 2017.

Publications

Minkyu Kim, Baekseung Kim, Junhoo Lee, Jangho Kim, Nojun Kwak, FuseBatch: Unlocking the Potential of Diffusion Models in Throughput Perspective, *Under Review*.

Junhoo Lee, Kyomin Hwang, Dongkwan Lee, Sangbum Han, **Minkyu Kim**, Nojun Kwak, Anchored eXtrapolation: Learning Framework without Human Priors, *Under Review*

Hyeonjin Kim, **Minkyu Kim**, Jae Jang, KiYoon Yoo, Nojun Kwak, TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Dermatological Diagnosis, *Proceedings of the 6th Clinical Natural Language Processing Workshop*, 2024.

Minkyu Kim, Suyoung Kim, Hyeonjin Kim, Sunghyun Wee, Junhoo Lee, Baekseung Kim, Nojun Kwak, A Marine Safety Management Model Using Drones, *36th Workshop on Image Processing and Image Understanding(IPIU)*, 2024.

Ho Seon Choi, **Minkyu Kim**, Geunwoo Shin, Junki Kim, Inyoung Kim, Yoon Su Baek, Development of the Algorithm for Controlling the Electric Walking Aids on the Slope by Using FSR Foot Sensor, *Journal of the Korean Society for Precision Engineering*, 2017.

Awards and Honors

Cum laude, Seoul National University	Aug 2024
Outstanding BS Thesis Presentation Award, Dept. of Mechanical Engineering, Seoul National University	June 2024
Mechanical Engineering Alumni Scholarship, Dept. of Mechanical Engineering, Seoul National University	Oct 2020
Bronze prize at Mechatronics Competition, Hyundai Infracore & Seoul National University	June 2023
Intel International Science and Engineering Fair 2017, South Korea National Delegation	May 2017

Military Service

The Republic of Korea's Army(ROKA) South Korea
The 50th Infantry Division Jan 2021 - July 2022

- Served as Sergeant, Squad Leader and Peer Counselor
- Performed duty as Operational Soldier and Duty Officer

Other Services

Captain of the Engineering College Basketball Team, Seoul National University	2024
Mentor of "KOTER: Catch-up Orientation Program", Seoul National University	2022
Vice-President of the Mechanical Engineering Student Council, Seoul National University	2020
Volunteer Member of "Visiting Engineering Volunteer Group," Seoul National University	2019
Welfare Division Member of the College of Engineering Student Council, Seoul National University	2019